

Qns	Answer	Marks
4(a)(i)	product of number of turns of coil, (magnetic) flux density and area area perpendicular to the (magnetic) field OR Magnetic flux linkage through a loop is product of magnetic flux through the loop and number of turns of wire in the loop. Magnetic flux is the product of an area and component of magnetic flux density perpendicular to that area.	M1 A1 B1 B1
4(a)(ii)	flux = $B \times \pi r^2$ = $0.17 \times \pi 0.36^2$ = 6.9×10^{-2} Wb time for one revolution = 1 / 25 s e.m.f. = rate of cutting flux or $\frac{\Delta\Phi}{\Delta t}$ = 0.069×25 = 1.7 V	C1 C1 A1
4(a)(iii)	current (in disc) is perpendicular to magnetic field causes force to act on disc OR current (in disc) is perpendicular to magnetic field OR current causes force to act on disc force opposes rotation of disc Fleming's left-hand rule indicates current is from rim to axle	B1 B1 B1

Qns	Answer	Marks
4(b)(i)	ring cuts (magnetic) flux and causes induced e.m.f. in ring (induced) e.m.f. causes (eddy/induced) currents (in ring) currents (in ring) cause magnetic field (around ring) two fields interact to cause resistive/opposing force OR current (in ring) is in a magnetic field which causes resistive force OR currents (in ring) dissipate thermal energy (thermal) energy comes from energy of oscillations	B1 B1 M1 A1 (M1) (A1) (M1) (A1)
4(b)(ii)	current cannot pass all the way around the ring (induced) currents smaller (OR no currents) smaller resistive force (so more oscillations) (OR no resistive force) or smaller rate of dissipation of energy (so more oscillations)	B1 B1 B1

